



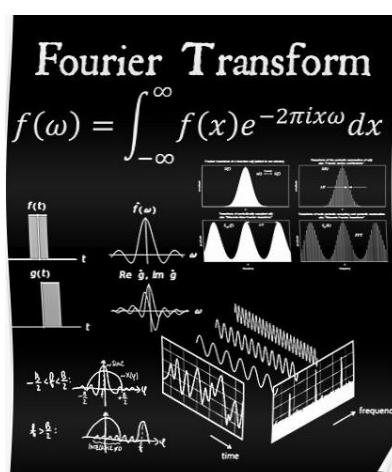
REDHILL SCHOOL



Tour de Maths 2023

Leg 5

Solutions



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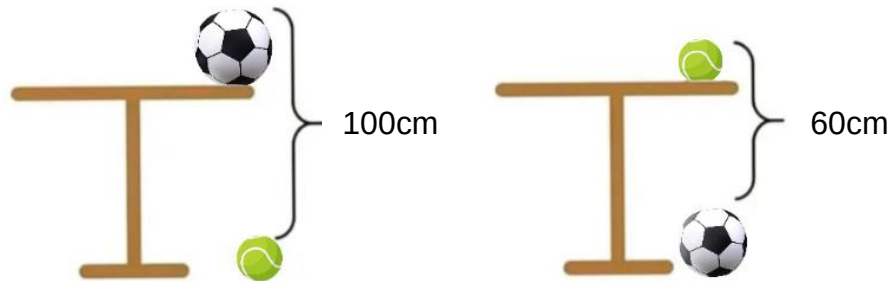


PRESH TALWALKAR

5 MARK QUESTIONS

QUESTION 1

What is the height of the table shown in the image below?



Let the height of the table be z .

Let the height of the soccer ball be x .

Let the height of the tennis ball be y .

From the left image: $z - y + x = 100$

From the right image: $z - x + y = 60$

Solving simultaneously: $z = 80$

Therefore, the height of the table is 80cm.

Answer: 80cm

QUESTION 2

If $a \oplus b = ab + \frac{a}{b}$, determine the value of $6 \oplus 2$.

$$6 \times 2 + \frac{6}{2} = 15$$

Answer: 15

QUESTION 3

Find the number formed by the last three digits of the sum

$$2! + 4! + 6! + 8! + \dots + 2018!$$

Where for a positive integer n , $n!$ is “ n factorial” which is the product of the integers from 1 to n , i.e. $1 \times 2 \times 3 \times 4 \times \dots \times n$.

All factorials from $16!$ have 4, 5, 10 and 15 as factors, whose product is 3000. Meaning from $16!$ onwards numbers will end in 3 zeroes.

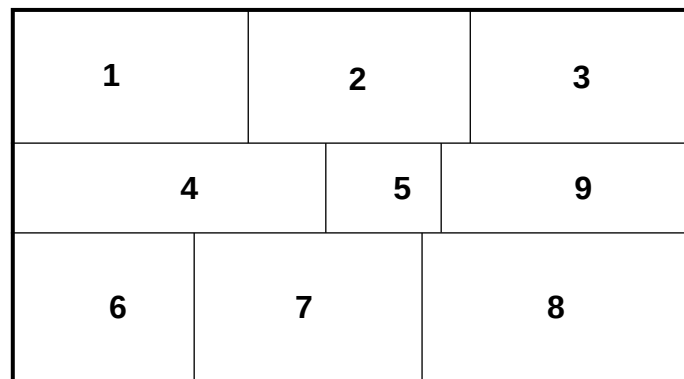
Meaning we only need to consider the sum of:

$$\begin{aligned} &2! + 4! + 6! + 8! + 10! + 12! + 14! \\ &= 2 + 24 + 720 + \dots 320 + \dots 800 + \dots 600 + \dots 200 \\ &= \dots 666 \end{aligned}$$

Answer: 666

QUESTION 4

In the diagram, a large rectangle is divided into nine smaller rectangles which are labelled 1 to 9, as shown below.



In how many ways can exactly two of these nine rectangles be shaded so that the shaded rectangles are not touching?

1-2					<i>KEY:</i>	Success	
1-3	2-3					Failure	
1-4	2-4	3-4					
1-5	2-5	3-5	4-5				
1-6	2-6	3-6	4-6	5-6			
1-7	2-7	3-7	4-7	5-7	6-7		
1-8	2-8	3-8	4-8	5-8	6-8	7-8	
1-9	2-9	3-9	4-9	5-9	6-9	7-9	8-9

Answer: 20

QUESTION 5

In how many ways can 22 be written as the sum of 3 different prime numbers? That is, determine the number of triples $(a; b; c)$ of prime numbers with $1 < a < b < c < 23$ and $a + b + c = 23$.

Every prime number other than 2 is odd. Since a, b and c are all prime and $a + b + c = 22$, which is even, it cannot be the case that all of a, b and c are odd (otherwise $a + b + c$ would be odd). Thus, at least one of a, b and c is even.

Since $1 < a < b < c < 23$, then it must be the case that $a = 2$ and b and c are odd primes. Since $a = 2$, then $b + c = 20$.

Since b and c are primes with $b < c$, then $b = 3$ and $c = 17$, or $b = 7$ and $c = 13$.

Therefore, there are two triples $(a; b; c)$ of primes numbers that satisfy the requirements.

Answer: 2 [i.e. $(2; 3; 17)$ and $(2; 7; 13)$]

QUESTION 6

Two people, starting at the same point, are running in opposite directions around a 400-metre circular path for 10 minutes.

If the speed of person A is 4 metres per second and the speed of person B is 5 metres per second.

How many times will person A and B pass each other?

By starting in opposite directions person 1 and person 2 will have run 400m when they cross each other.

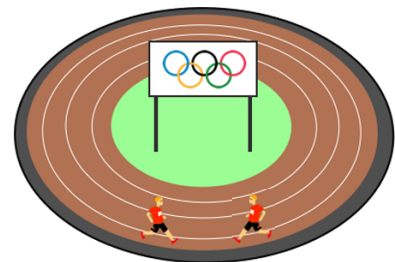
Let the distance covered by person 1 be x and person 2 be $400 - x$.

$$\text{Person 1: } 4 = \frac{x}{t} \quad \text{Person 2: } 5 = \frac{400 - x}{t}$$

Solving simultaneously $t = \frac{400}{9}$ meaning in 10 minutes (600 seconds) they will cross

each other $\left(\frac{600}{\frac{400}{9}}\right) = 13,5$ times which rounds to 13 times.

Answer: 13 times



QUESTION 7

How many numbers are there in the list 1; 2 ; 3 ; 4 ; ; 10000 which contain exactly two consecutive 9's such as 993, 1992 and 9929 but not 9295 or 1999?

Consider 4 digits and include numbers written as 0020 (20).

There are three possible positions for 99: 99xy OR x99y OR xy99

$$99xy: 9 \times 10 = 90$$

$$x99y: 9 \times 9 = 81$$

$$xy99: 10 \times 9 = 90$$

$\therefore$ 261 numbers

Answer: 261

QUESTION 8

Suppose $X = 1 + 11 + 101 + 1001 + 10001 + \dots + \overbrace{1000\dots0001}^{50 \text{ ZEROES}}$

When X is calculated and written as a single integer, the sum of its digits is?

- (A) 58 (B) 99 (C) 52 (D) 50 (E) 103

There are 52 terms in the sum: the number 1, the number 11, and the 50 numbers starting with a 1, ending with a 1 and with 1 to 50 zeroes in between.

The longest of these terms thus has 52 digits (50 zeroes and 2 ones).

When the unit's digits of all 52 terms are added up, their sum is 52, so the units digit of X is 2, and a 5 carried to the tens digit.

In the tens digit, there is only 1 non-zero digit: the 1 in the number 11. Therefore, using the carry, the tens digit of X is $1 + 5 = 6$. In each of positions 3 to 52 from the right-hand end, there is only one non-zero digit, which is a 1.

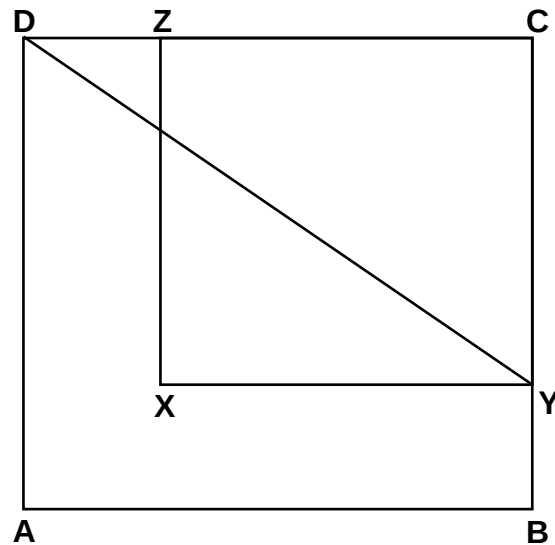
Therefore, the digit in each of these positions in X is also a 1. (There is no carrying to worry about.)

Therefore, $X = 11 \dots 1162$, where X has $52 - 50 = 2$ digits equal to 1. This tells us that the sum of the digits of X is $50(1) + 6 + 2 = 58$.

Answer: A

QUESTION 9

ABCD and XYCZ are squares. The area inside square ABCD, but outside square XYCZ is 30 square units.



Determine the length of CD if $DY = 10$ units.

$$x^2 = y^2 + 30 \quad \& \quad DY^2 = x^2 + y^2$$

$$\therefore 100 = x^2 + y^2$$

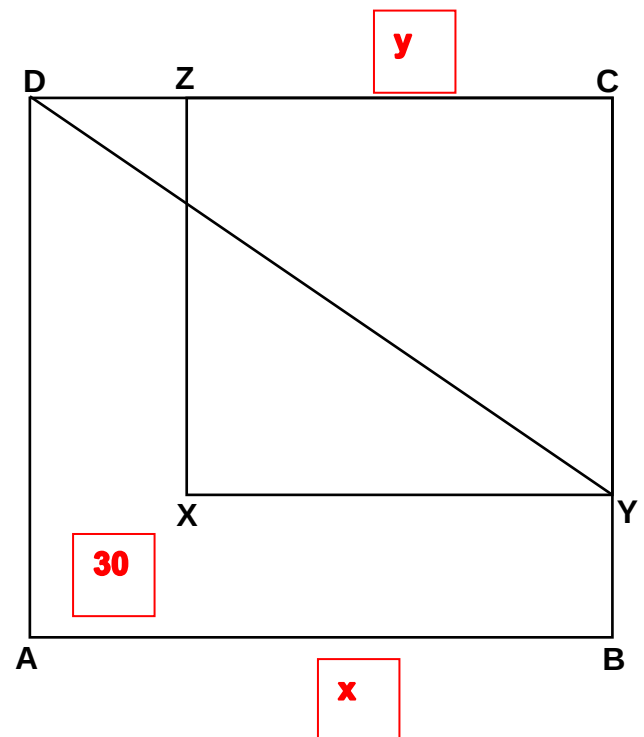
$$\therefore x^2 = 100 - x^2 + 30$$

$$\therefore 2x^2 = 130$$

$$\therefore x^2 = 65$$

$$\therefore x = \sqrt{65}$$

Answer: $\sqrt{65}$



QUESTION 10

Find the unit digit of $5^{100} - 2^{100}$.

- (A) 5 (B) 0 (C) 2 (D) 4 (E) 9

Let us observe the periodic nature of the unit digit of the powers of 5.

$5^1 = 5$. Unit digit is 5.

$5^2 = 25$. Unit digit is 5.

$5^3 = 125$. Unit digit is 5.

Thus, 5^n will have a unit digit of 5.

5^{100} will have a unit digit of 5.

Now, let us observe the periodic nature of the unit digit of the powers of 2.

$2^1 = 2$. Unit digit is 2.

$2^2 = 4$. Unit digit is 4.

$2^3 = 8$. Unit digit is 8.

$2^4 = 16$. Unit digit is 6.

$2^5 = 32$. Unit digit is 2.

This pattern will repeat.

How many iterations fit into 100?

That is $\frac{100}{4}$ which is 25. Thus 25 full interactions fit into 100. Therefore, we reach

the end of the end of the pattern, reaching 6. The unit digit of 2^{100} is therefore 6.

Now, $5 - 6 = -1$ and -1 cannot be a unit digit.

Therefore, 10 will be subtracted from the tens unit and added to the unit digit resulting in 9.

Answer: E

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QUESTION 11

A certain school is throwing a dance. The school council want to give a welcome gift to each student that attends the dance.

Gifts-R-Us charges R1,00 per gift. However, if they were to purchase the gifts at *Gifts-R-Us*, they would exceed their budget by R17.

At *Presents-4-U*, they only charge R0,80 per gift. At this price, the student council would have R5 left over in their budget.

Determine the number of gifts the student council is planning to buy.

Let n represent the number of gifts that the student council is planning to buy.

Since each gift at Gifts-R-Us costs R1.00, the student council would spend $1 \times n = n$ rands in total.

If the student council were to purchase all of the gifts they want at Gifts-R-Us, they would be short R17 dollars in their budget. Therefore, the amount they have in their budget is $R(n - 17)$.

Since each gift at Presents-4-U costs R0.80, the student council would spend $0,8 \times n = 0,8n$ rands in total.

If the student council were to purchase all of the gifts they want at Presents-4-U, they would have R5 left over in their budget.

Therefore, the amount they have in their budget is $(0,8n + 5)$ rands. We have two expressions for the amount in their budget, so we can establish the equality $n - 17 = 0,8n + 5$. This simplifies to $0,2n = 22$. After dividing each side by 0,2, we obtain $n = 110$. Therefore, the student council is planning to buy 110 gifts.

Answer: 110

QUESTION 12

If it is given that $a+1=b+2=c+3=d+4=a+b+c+d+5$, what is the value of $a+b+c+d$?

$$(a+1)+(b+2)+(c+3)+(d+4)=4x$$

$$(a+b+c+d)+10=4x$$

$$(a+b+c+d+5)+5=4x$$

$$x+5=4x$$

$$3x=5$$

$$x=\frac{5}{3}$$

$$a+b+c+d+5=\frac{5}{3}$$

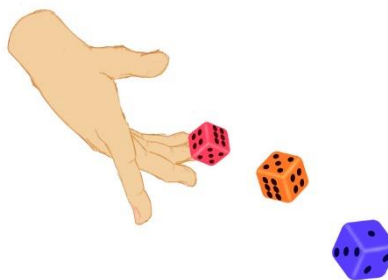
$$a+b+c+d=\frac{5}{3}-5$$

$$a+b+c+d=-\frac{10}{3}$$

Answer: $-\frac{10}{3}$

QUESTION 13

Three, fair, six-sided dice are thrown.



What is the probability that there sum is 9?

Total outcomes: 216

The combinations that have a sum of 9 are:

126, 135, 144, 153, 162, 216, 225, 234, 243, 252, 261, 315, 324, 333, 342, 351,

414, 423, 432, 441, 513, 522, 531, 612, 621

The probability of getting a sum of 9 is $\frac{25}{216}$ or approximately 11.574%.

Answer: $\frac{25}{216}$

QUESTION 14

A number is called **special** if it is the sum of a non-negative multiple of 4 and a non-negative multiple of 7.

So, 11 and 12 are special since $11 = 1 \times 4 + 1 \times 7$ and $12 = 3 \times 4 + 0 \times 7$ but 13 is not special.

How many special numbers are there in the interval 1 to 1000?

The key observation is that since we have the 4 consecutive special numbers,

$$18 = 1 \times 4 + 2 \times 7$$

$$19 = 3 \times 4 + 1 \times 7$$

$$20 = 5 \times 4 + 0 \times 7$$

$$21 = 0 \times 4 + 3 \times 7$$

All larger numbers are special, because they can be obtained by adding a multiple of 4 to one of 18, 19, 20 or 21.

This leaves determining which numbers less than 18 are special.

The restriction, $n \times 4 + m \times 7 < 18$ implies $m \leq 2$.

Checking, we find the special numbers less than 18 are: for $m = 0$: 4, 8, 12 and 16, for $m = 1$: 7, 11 and 15, for $m = 2$: 14.

Since, in this search, we did not encounter 1, 2, 3, 5, 6, 9, 10, 13 or 17 (9 of them), these numbers are not special, and they are the only non-special numbers less than 1000.

Thus, in all, there are $1000 - 9 = 991$ special numbers in the interval 1 to 1000.

Answer: 991

QUESTION 15

In Miss Q's class $\frac{1}{3}$ of the students brought a teddy bear to school.

Each boy in the class took 12 books, each girl took 17 books, and each teddy bear took 9 books.



A total of 305 books were taken out.

How many girls were there in Miss Q's class?

If $\frac{1}{3}$ brought teddy bears then the total must be divisible by 3.

The smallest value that will work within these parameters is 18.

Therefore boys + girls = 18 and $12\text{boys} + 17\text{girls} + 54 = 305$.

Therefore, there are 11 boys and 7 girls.

Answer: 7 girls

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QUESTION 16

An arithmetic (also known as linear) sequence is a sequence in which each term after the first is obtained by adding a constant to the previous term.

For example: 4, 7, 10, 13, 16 is an arithmetic sequence with five terms, in which the first term is 4 and the constant added is 3.

Each row and each column in the 5 x 5 array below is an arithmetic sequence with five terms.

1				17
	A			
25				81

Calculate the value of A .

1		9		17
13	A = 22	31		
25		53		81

$$\frac{1+17}{2} = 9$$

$$\frac{25+81}{2} = 53$$

$$\frac{9+53}{2} = 31$$

$$\frac{1+25}{2} = 13$$

$$\frac{31+13}{2} = 22$$

Answer: A = 22

QUESTION 17

In the diagram below, a positive integer is hidden behind each star.

The integer shown on each star is the product of the integers hidden behind the other three stars.



What is the product of all four hidden integers?

Suppose that the numbers hidden behind the four stars, in order, are w ; x ; y ; z .

Since the number on the first star is 280, then the product of the numbers behind the other three stars is 280, or $xyz = 280$.

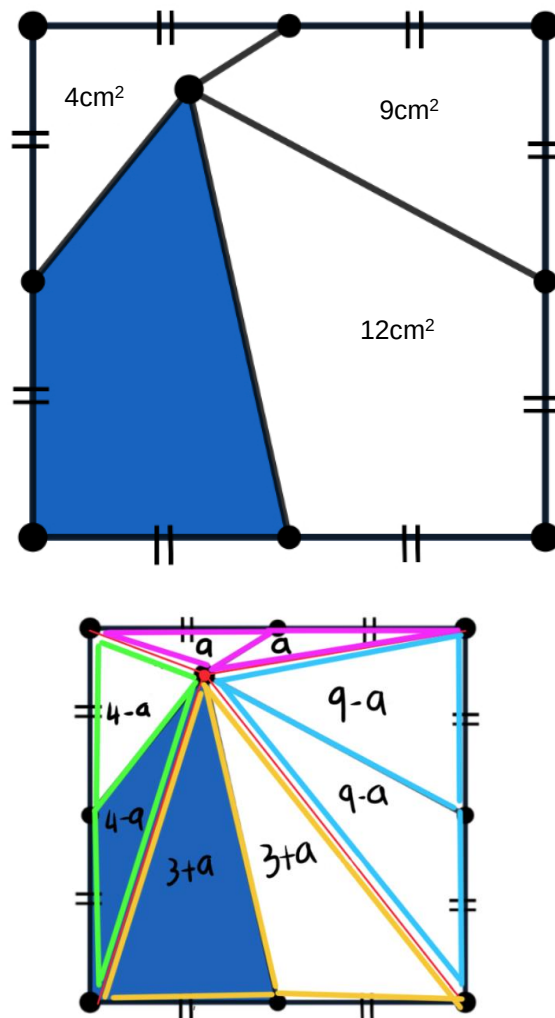
Similarly, the numbers on the other three stars give the equations $wyz = 168$, $wxz = 105$ and $wxy = 120$.

Multiplying these four equations together, we obtain

$$\begin{aligned}(xyz)(wyz)(wxz)(wxy) &= (280)(168)(105)(120) \\ w^3 \cdot x^3 \cdot y^3 \cdot z^3 &= (2^3 \cdot 5 \cdot 7)(2^3 \cdot 3 \cdot 7)(3 \cdot 5 \cdot 7)(2^3 \cdot 3 \cdot 5) \\ (wxyz)^3 &= 2^9 \cdot 3^3 \cdot 5^3 \cdot 7^3 \\ (wxyz)^3 &= (2^3 \cdot 3 \cdot 5 \cdot 7)^3 \\ wxyz &= 2^3 \cdot 3 \cdot 5 \cdot 7 \\ wxyz &= 840\end{aligned}$$

QUESTION 18

Determine the area of the blue shaded region below.



Therefore, the blue area = $4 - a + 3 + a = 7$

Answer: 7cm^2

QUESTION 19

Twelve squares are placed in a row forming the grid below. Each square is to be filled with an integer. After the third square, each integer in a square is the sum of the previous three integers. If we know the third integer is 6, the sixth integer is 11, and the eleventh integer is 14.

		6		Q	11					14	
--	--	---	--	-----	----	--	--	--	--	----	--

Determine the value of Q .

Let a_1 be the first integer in the grid, a_2 be the second, a_3 be the third, a_4 be the fourth, and so on, until a_{12} which is the twelfth integer in the grid.

We are given that $a_3 = 6$, $a_6 = 11$, and $a_{11} = 14$.

Each integer after the third integer is equal to the sum of the previous three integers.

Therefore, $11 = 6 + a_4 + a_5 \therefore a_4 + a_5 = 5$.

We also have $a_7 = a_4 + a_5 + a_6 = a_4 + a_5 + 11 = 5 + 11 = 15$ since $a_4 + a_5 = 5$.

Similarly, $a_9 = a_6 + a_7 + a_8 = 11 + 15 + a_8 = 26 + a_8$,

$$a_{10} = a_7 + a_8 + a_9 = 15 + a_8 + a_9 = 15 + a_8 + (a_8 + 26) = 2a_8 + 41,$$

$$\text{and, } a_{11} = a_8 + a_9 + a_{10} = (a_8) + (a_8 + 26) + (2a_8 + 41) = 4a_8 + 67.$$

We are given that $a_{11} = 14$. Therefore, $4a_8 + 67 = 14$, or $4a_8 = -53$ or $a_8 = -13.25$.

Therefore, $a_9 = a_8 + 26 = -13.25 + 26 = 12.75$, and $a_{10} = 2a_8 + 41 = 15.5$.

Also, $a_{12} = a_9 + a_{10} + a_{11} = 12.75 + 15.5 + 14 = 42.25$.

Working backwards, $-14 = a_5 + 11 + 16$ so $a_5 = -41$.

Thus $Q = -41$

Answer: -41

QUESTION 20

Noah and Acacia each have nine cards, numbered from 1 to 9. Each person randomly discards one of their cards and then calculates the sum of the numbers on their remaining eight cards.



Calculate the probability that the difference between these two sums is a multiple of 4.

If Noah discarded the card numbered 1, the sum of the numbers on the remaining cards would be $45 - 1 = 44$, which is n .

If Acacia also discarded the card numbered 1, then $a = 44$ as well. Then $n - a = 0$, which is a multiple of 4. Acacia could also discard the cards numbered 5 or 9.

Then $a = 45 - 5 = 40$ or $a = 45 - 9 = 36$, and $n - a = 4$ or $n - a = 8$, both of which are multiples of 4.

In general, if Acacia discards a card with the same number as Noah or the same number as Noah increased or decreased by some multiple of 4, then $n - a$ will be a multiple of 4.

We find that there are $3 + 2 + 2 + 2 + 3 + 2 + 2 + 2 + 3 = 21$ different ways to discard the cards so that n and a differ by a multiple of 4.

Therefore, the probability that n and a differ by a multiple of 4 is $\frac{21}{81} = \frac{7}{27}$.

Answer: $\frac{7}{27}$